

WHAT IS IQR?

DEFINITION:

The Interquartile Range (IQR) is a measure of statistical dispersion that represents the range within which the middle 50% of the data lies.

It is calculated as the difference between the third quartile (Q3) and the first quartile (Q1).

FORMULA:

$$\text{IQR} = Q3 - Q1$$

INTERQUARTILE RANGE (IQR)

Why Use $1.5 \times \text{IQR}$ for Outlier?

As per Tukey's Rule introduced by American statistician John Wilder Tukey in 1977 in his book Exploratory Data Analysis, outliers are detected using Lower Bound = $Q1 - 1.5 \times \text{IQR}$ and Upper Bound = $Q3 + 1.5 \times \text{IQR}$.

The value 1.5 is not a strict mathematical law but an empirical standard; in a normal distribution this fence covers about 99.3% of data (around 2.7 sigma),

so values outside are considered rare and marked as mild outliers, while $3 \times \text{IQR}$ is used for extreme outliers. **1.5 is chosen as a balanced threshold because $1 \times \text{IQR}$ gives too many false outliers and $2 \times \text{IQR}$ misses real outliers.**

EXAMPLE FOR IQR

- a. The interquartile range. Compare the two interquartile ranges.
- b. Any outliers in either set.

The five number summary for the day and night classes is

	Minimum	Q1	Median	Q3	Maximum
Day	32	56	74.5	82.5	99
Night	25.5	78	81	89	98

Solution:

a) IQR Calculation

Formula:

$$\mathbf{IQR = Q3 - Q1}$$

Day :

$$IQR = 82.5 - 56 = 26.5$$

Night :

$$IQR = 89 - 78 = 11$$

Comparison:

The middle 50% of the Day class scores are spread over a range of 26.5, while the middle 50% of the Night class is spread over only 11. Therefore, the Day class has more variation, while the Night class is more consistent. Most Night class students scored between 78 and 89.

b) Are there any outliers?

Formula:

$$\text{Lower Bound} = Q1 - 1.5 \times IQR$$

$$\text{Upper Bound} = Q3 + 1.5 \times IQR$$

For Day Class:

$$\text{Lower} = 56 - 1.5 \times 26.5$$

$$= 56 - 39.75 = \mathbf{16.25}$$

$$\text{Upper} = 82.5 + 39.75$$

$$= \mathbf{122.25}$$

$$\text{Min (32)} > 16.25 \text{ and } \text{Max (99)} < 122.25$$

=> No outliers in Day class.

For Night Class:

$$\text{Lower} = 78 - 1.5 \times 11 = 78 - 16.5 = \mathbf{61.5}$$

$$\text{Upper} = 89 + 16.5 = \mathbf{105.5}$$

$$\text{Min (25.5)} < 61.5$$

CONCLUSION:

There is 1 outlier in Night class. The value 25.5 is a low outlier.

There is no high outlier.

The score of 25.5 is significantly lower compared to the rest of the Night class.